

NOMAD : New Operations & Market Advisory Director

What Is It?

NOMAD is an AI driven advisory system that helps businesses make better decisions about their physical operations such as locations, facilities, suppliers, warehouses, and expansion plans. It acts as a digital operations partner understanding a company's physical footprint and recommending what to do next.

It is built on **Mireye**, a physical world intelligence layer covering demographics, zoning, permits, hazards, and accessibility. This is combined with business system data and public/real time sources, then processed through an AI reasoning layer to produce site rankings, feasibility reports, risk flags, compliance checks, and logistics plans.

Initial capabilities include, address and facility verification, site evaluation, regulatory research, risk monitoring for facilities and routes, multi location comparison, expansion site selection, identifying where local exceptions are needed, catchment based demand estimation, and reverse logistics/inventory transfer planning.

It isn't meant to replace managers or consultants, it's meant to give them a more reliable decision support layer as the business scales from one location to many, surfacing regional differences and operational risks that standard playbooks miss.

Rather than being a collection of separate tools, it acts as a continuous business partner that can analyze a company's locations and answer questions such as:

1. How should our physical network be optimized?
2. What regulations apply?
3. What risks exist around our locations?
4. Is this facility suitable?
5. Where should we expand?

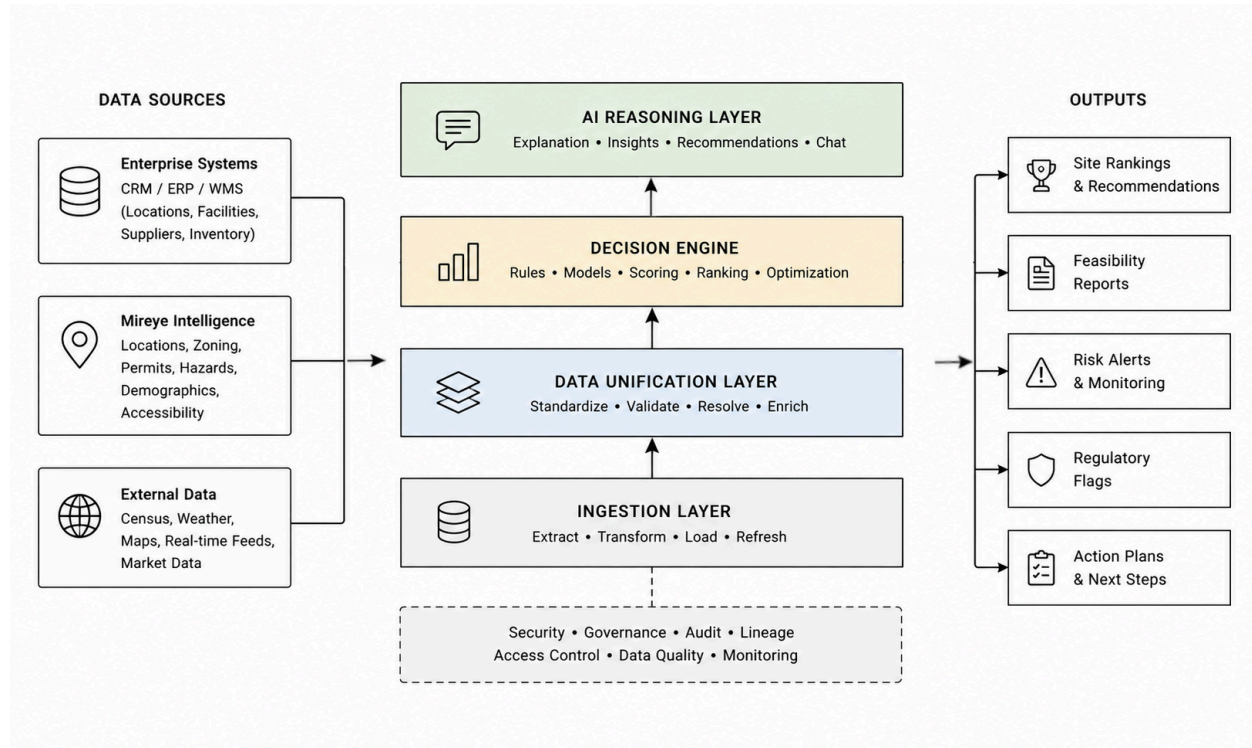
Product Positioning

Supports businesses across four roles:

1. **AI Consultant:** Recommends the best decisions by evaluating locations and physical world factors.
2. **Planner:** Converts analysis into actionable plans for expansion, logistics, and operations.
3. **Analyst:** Continuously monitors locations, compares changes, and identifies emerging risks.
4. **Operational Partner:** Connects physical world intelligence with business systems to support ongoing decisions and local exceptions.

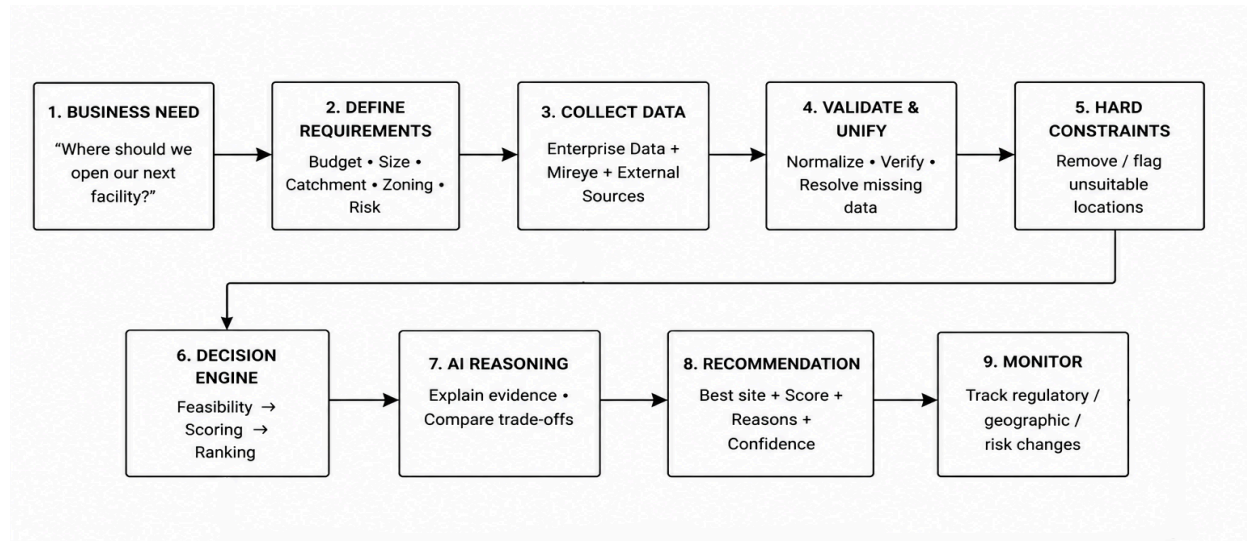
A client could give it their locations, facilities, suppliers, warehouses, and expansion plans, and continuously ask it to analyze, compare, verify, monitor, and recommend.

System Architecture



1. **AI Reasoning Layer:** The top level intelligence that processes unified physical world data and turns it into recommendations, rankings, and decisions through a natural language and decision engine.
2. **Mireye Intelligence Layer:** The core physical world intelligence layer that standardizes information such as locations, zoning, permits, hazards, demographics, and accessibility.
3. **Data Source Layer:** The foundation that brings together enterprise data from systems such as CRM, ERP, and WMS, alongside Mireye data and external sources such as census, weather, and real time feeds.
4. **Output Layer:** Converts the analysis into business ready outcomes such as feasibility reports, risk alerts, site rankings, logistics plans, and compliance flags.

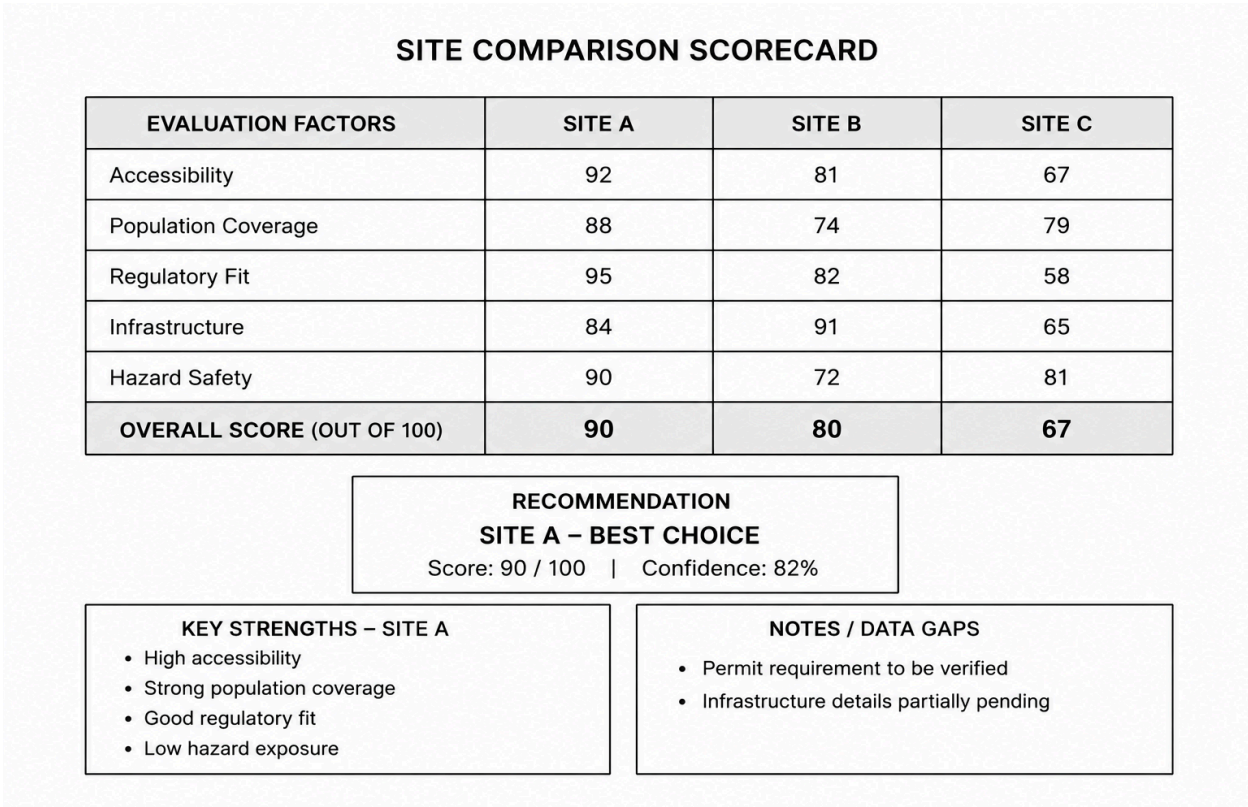
Data Flow Model



Nine Step Business Decision Flowchart

1. **Recommend:** Business recommendations and actions are generated: feasibility reports, risk alerts, site rankings, logistics plans
2. **Analyze:** The AI reasoning engine processes unified data to identify patterns, risks, opportunities, and trade offs
3. **Unify:** The Enterprise Physical Data Unification Layer standardizes and exposes all physical world information consistently
4. **Ingest:** Enterprise systems (CRM, ERP, WMS), Mireye physical-world data, and external public data feeds are ingested

Core Capabilities



Site Comparison Scorecard and Recommendation (Not Real)

1. Enterprise Physical Data Unification Layer

Standardize and expose physical world information for the company’s AI and business systems. This is the foundational layer that makes all other capabilities possible.

2. Site & Facility Feasibility Screening

Determine whether candidate facilities/sites satisfy physical and geographic requirements. Screens locations against criteria such as accessibility, zoning, infrastructure, and environmental factors.

3. Regulatory, Zoning & Hazard Data Layer

Monitor zoning, permits, hazards, and other location level risks. Enhanced with live external feeds for real time regulatory and hazard updates.

4. Location Based Regulatory & Permitting Research

Investigate location specific zoning and permitting requirements. Provides detailed research on what permits are needed, what zoning applies, and what restrictions exist for a given address.

5. Address & Facility Verification

Verify that addresses correspond to legitimate, appropriately classified physical locations. Prevents errors from invalid, misclassified, or non-existent addresses in enterprise systems.

6. Facility & Route Risk Monitoring

Identify geographic risks affecting facilities, suppliers, and routes. Enhanced with live external feeds for weather, natural disasters, and other real time risk factors.

7. Multi Location Operations Monitor

Compare and monitor the physical/regulatory context of an entire location network. Provides a dashboard view across all company locations for comparative analysis.

8. Facility & Site Selection

Rank potential locations based on physical, geographic, regulatory, and demographic suitability. The flagship decision support capability for expansion planning.

9. Central vs. Local Decision Engine

Identify where standardized company decisions may require local exceptions because of geographic differences. Helps balance corporate standardization with local compliance and optimization.

10. Catchment Based Population & Demand Engine

Analyze surrounding populations and use external free/public data to estimate geographic demand. Combines Mireye demographic data with census and other public sources.

11. Reverse Logistics & Returns Network Planner

Determine geographically suitable destinations for returns, repairs, refurbishment, recycling, and disposal. Optimizes the reverse supply chain based on physical world factors.

12. Multi Location Inventory Transfer Planner

Use geographic relationships and accessibility to optimize potential inventory transfers between locations. Considers distance, accessibility, regulatory constraints, and risk.

Future Product Additions

Once the core platform is established, additional capabilities can be added as modules. These should be treated as future extensions.

1. **Supplier Geographic Concentration Mapping:** Supplier clustering risks
2. **Location Aware Workforce Planner:** Location based staffing
3. **Distribution Site & Competitor Proximity Mapping:** Competitor proximity
4. **Market Sizing:** Demographic & economic market sizing
5. **Regional Customer Service Context Engine:** Location-aware customer support
6. **Facility Verification:** Facility & certification verification
7. **Labor Market Sizing by Location:** Labor availability, costs & skills

Two Week Plan

Vedant Awasthi will lead backend development and Mireye integration, including data ingestion, normalization, API connectivity, and the unified physical world data layer.

Shaurya Suyal will lead the AI and decision engine, focusing on reasoning, scoring, ranking, recommendations, and capability specific decision models.

Paarth Gupta will lead the frontend and product experience, including the dashboard, visualization, user workflow, demo experience, and supporting documentation.

Week 1: Foundation and Core Integration

The first two days will establish the technical foundation. **Vedant** will define the architecture, repository, module boundaries, unified data schema, and Mireye integration requirements.

Shaurya will define the AI reasoning and recommendation pipeline, including initial location scoring and ranking. **Paarth** will establish the dashboard structure, user journey, project board, and sprint tracking.

From **August 17 to 18**, the team will build the enterprise physical world data layer. Vedant will develop ingestion, normalization, validation, and CSV/JSON connectors. Shaurya will define the AI representation of standardized location data, prepare reasoning prompts, and establish the backend AI interface. Paarth will build the dashboard shell, location import, location profile, and recommendation views.

On **August 19**, the focus shifts to Mireye integration. Vedant will implement the API client, authentication, and data mapper, Shaurya will connect Mireye derived data to the AI reasoning layer and run initial tests and Paarth will connect the dashboard to the backend and display Mireye data.

Week 2: Capability Build and Validation

Week 2 will turn the foundation into working decision capabilities. From **August 22 to 24**, Vedant will build backend support for **Address & Facility Verification, Site & Facility Feasibility, and Regulatory & Zoning**, including the required AI and frontend endpoints.

Shaurya will develop the **Site Selection scoring**, feasibility scoring, and hazard interpretation using structured rules and measurable factors, with AI layered on top for explanations and recommendations. Paarth will turn these into dashboard workflows and add a basic **Multi Location Operations Monitor** for comparison and monitoring.

On **August 25 and 26**, the focus shifts to validation. Vedant will test the complete ingestion pipeline, representative addresses, API failures, and missing or incomplete data. Shaurya will validate feasibility and site selection models against known good/bad examples, check ranking consistency, and test AI outputs for hallucinations, unsupported claims, and contradictions. Paarth will test the full **location input to recommendation** workflow, improve usability and information hierarchy, and fix remaining frontend issues.